Mid Semester Report on the Study of Quasi Normal Modes in $f(R)$ Gravity

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1. Introduction

The study of quasi-normal modes (QNMs) in $f(R)$ gravity has gained considerable attention in recent years. These modes characterize the ringdown phase of black hole perturbations and provide a potential avenue for testing deviations from General Relativity (GR) through gravitational wave observations.

My project aims to analyze the QNM spectra of black holes in various $f(R)$ models, comparing them to their GR counterparts. A crucial aspect of this study is to investigate QNMs in the conformally related Einstein frame, where the additional degree of freedom in $f(R)$ gravity—the scalaron—appears explicitly as a scalar field coupled to gravity. This approach allows us to understand how metric perturbations interact with scalar field perturbations and what role they play in the **stability** of black holes in these theories.

2. Progress So Far

2.1 Understanding $f(R)$ Gravity and Its Einstein Frame Representation

One of the first challenges was to fully understand the formulation of $f(R)$ gravity and its transformation to the Einstein frame. This required a careful study of:

- The field equations in the Jordan frame and their modifications compared to Einstein's equations in GR.
- The transformation via a conformal rescaling to obtain the Einstein frame representation.
- The emergence of a scalar field (the scalaron) from the extra degrees of freedom in $f(R)$ gravity.

After several attempts, I successfully derived the transformation equations and understood their implications. The key result is that under the metric transformation $\tilde{g}_{\mu\nu} = \phi g_{\mu\nu}$, with $\phi = f'(R)$, the action in the Einstein frame takes the form

$$S = \int d^4x\sqrt{-\tilde{g}}\left(\tilde{R} - \frac{1}{2}\left(\tilde{\nabla}\tilde{\phi}\right)^2 - U\left(\tilde{\phi}\right)\right) + S_M,$$

where ϕ is related to $\tilde{\phi}$ by $\tilde{\phi} = \sqrt{\frac{3}{2\kappa}}\ln\phi$. This is general relativity coupled with scalar field $\tilde{\phi}$.

2.2 Review of QNMs in General Relativity

Before analyzing QNMs in $f(R)$ gravity, I reviewed the standard procedure for computing QNMs in GR. This involved reviewing my previous project on QNMs of Schwarzschild black holes (from Rezzolla's lecture notes, and Kokkotas' Living Review in GR):

- **Perturbation theory in Schwarzschild spacetime**: Understanding the Regge-Wheeler and Zerilli equations for axial and polar perturbations.

- **Sturm-Liouville theory and eigenvalue problems**: Recognizing how QNMs appear as solutions to boundary value problems.
- **WKB approximation**: used to approximate analytical expressions of QNM frequencies.

This review helped solidify my understanding of black hole perturbation theory and provided a foundation for extending the analysis to $f(R)$ gravity.

2.3 Derivation of Field Equations in $f(R)$ Gravity

Having established the Einstein frame formulation, I focused on deriving the field equations for metric perturbations in $f(R)$ gravity. The key points of progress are:

- Deriving the modified Einstein equations in the Jordan frame.
- Understanding the conformal transformation and obtaining the Einstein frame field equations.
- Identifying the additional degrees of freedom introduced by $f(R)$ gravity.

At this stage, I have successfully obtained the field equations that govern gravitational perturbations in both frames. However, the explicit perturbation equations for black holes in $f(R)$ gravity still need to be worked out.

3. Next Steps

The next phase of the project involves:

1. **Deriving and solving the perturbation equations for black holes in $f(R)$ gravity**: This will involve separating axial and polar perturbations and analyzing their structure in both Jordan and Einstein frames.
2. **Computing the QNM spectrum**: Once the perturbation equations are established, I will explore techniques such as the WKB approximation and other methods.
3. **Comparing results in Jordan and Einstein frames**: This will provide insights into how the presence of the scalar field affects black hole stability and QNM spectra.
4. **Exploring physical implications**: I will analyze whether deviations from GR in QNMs could be observable in future gravitational wave detections.

4. Conclusion

So far, I have made significant progress in understanding $f(R)$ gravity, its Einstein frame formulation, and the derivation of field equations. While deriving the explicit perturbation equations and computing QNMs remains an ongoing challenge, I am now equipped with the necessary tools to tackle these problems. The upcoming work will focus on explicit computations to compare QNMs in different gravity models.

By the end of this project, I aim to provide a comprehensive comparison of QNMs in various $f(R)$ gravity models, highlighting their differences from GR and their potential observational significance.

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